



Economics AP Review Guide

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Range: AP Microeconomics

Note: (Review Guide/Review Practices/Practice Set) are UNOFFICIAL sources of reference and do NOT include or hint at questions in the actual exam!

Ch1 Principles of Economy:

Economics:

What is Economics:

How the society manage scarce resources

Scarcity: Limited nature of society's resources

Questions mainly asked:

What to buy, amount of work, save and spend

What to produce, employment→Dividing resources

- **Why is Economics a Science:**

- Collect data
- Use theories to explain phenomena
- Analyze: natural experiments offered by history
- Scientific method: testing of theories and observations

Microeconomics vs Macroeconomics

Microeconomics	Macroeconomics
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Study of how Households and firms make decisions	Study of economy-wide phenomena (How nation make decisions)
Ex: what to buy, how much to produce	Ex: Inflation, unemployment, growth

Positive vs Normative statements

Positive statements	Normative statements
Descriptive statement (fact)	Prescriptive statement (opinion)
Ex: The price is rising	Ex: Price will rise in 5 years

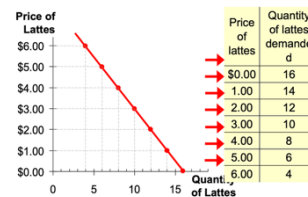
Principles of Economy:

1. People Face tradeoff:
 - Efficiency vs Quality
2. Cost of something = What you give up:
 - Opportunity Cost: second best alternative given up (inhibit+ exhibit)
3. Rational people think the margin:
 - Systematically do the best to profit
 - Compare Marginal benefit & Marginal cost
4. People Respond to Incentives:
 - Incentive: something that induces a person to act
 - a. Award
 - b. Punishment

Ch4 Demand and Supply:

Quantity Demanded (Qd):

- amount of the goods that buyers are willing and able to purchase.
- **Law of demand:** Qd of a good fall when the price of the good rises, other things equal
- **Demand schedule:** a table that shows the relationship between the price of a good and the Qd

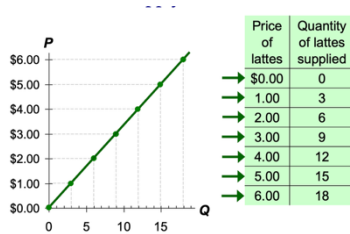


- Demand Curve shifters:
 - a. Increase in # of buyers → increases Qd
 - b. Income:
 - Normal goods: Increase in income → increase Demand (& vice versa)
 - Inferior goods: Increase in income → decrease Demand (& vice versa)
 - c. Substitutes/Complements:
 - Substitute: An increase price of one causes an increase in demand for the other
 - Complements: An increase price of one causes a fall in demand for the other.
 - d. Taste: Whether favor a good/ service over another or not.
 - e. Expectations: people expect incomes to rise → Demand for goods to increase now

Quantity Supplied (Qs):

- amount that sellers are willing and able to sell

- **Law of supply:** Qs of goods rise when the price of the good rises, other things equal
- **Supply schedule:** A table that shows the relationship between the price of a good and the quantity supplied.



- Supply curve shifters:
 - f. Input price: Higher input price → Decrease in S
 - g. Technology: Better Technology → Increase in S
 - h. Number of sellers: increase number → Increase in S
 - i. Expectations: adjust supply when their expectations of future prices change

Ch5 Elasticity:

Elasticity: the numerical measure of the responsiveness of Q^d or Q^s

Price elasticity of demand (PED):

- measures how much Q^d responds to a change in P

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in } Q^d}{\text{Percentage change in } P}$$

- Midpoint Method: Calculating PED

$$\frac{\text{end value} - \text{start value}}{\text{midpoint}} \times 100\%$$

- Determinants of PED:
 - j. Existence of Substitutes
 - k. Necessity or a Luxury
 - l. Defined Broadly or Narrowly
 - m. Long run or short run
- The higher the elasticity, the flatter the curve is

Price Elasticity of Supply (PES):

- how much Q^s respond to a change in P .

$$\text{Price elasticity of supply} = \frac{\text{Percentage change in } Q^s}{\text{Percentage change in } P}$$

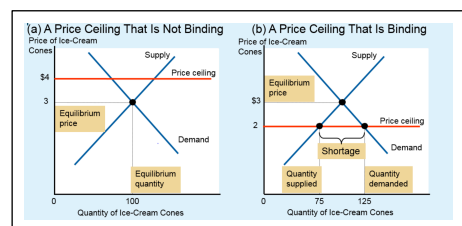
Ch6 Price Controls:

Price Control:

- **Purpose:**
 - Deal with the inequalities generate within the market

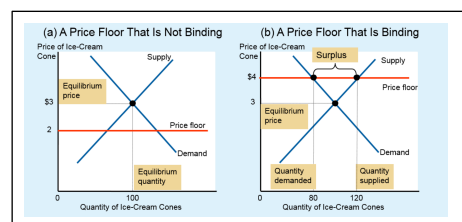
- **Price Ceiling:**

- legal maximum on the price
- ex: Rent-control laws



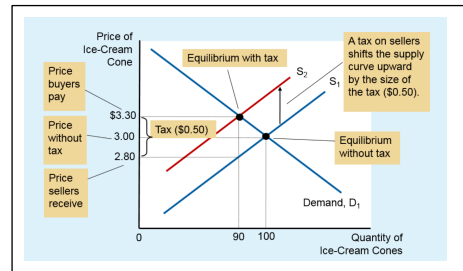
- **Price Floor:**

- legal minimum on the price
- ex: Minimum wage laws



- **Tax:**

- raise revenue for public purposes
- influence market outcomes



Ch7 Welfare Economy:

Welfare economics: the study of how the allocation of resources affects economic well-being

Economic well-being: a person's or family's standard of living based primarily on how well they are doing financially. (Can be calculated by GDP per capita.)

What determines the **allocation of resources**:

- how much of each good is produced
- which producers produce it
- which consumers consume it

1. Consumer Surplus (CS):

Amount a buyer is willing to pay minus the amount the buyer pays

2. Producer surplus (PS):

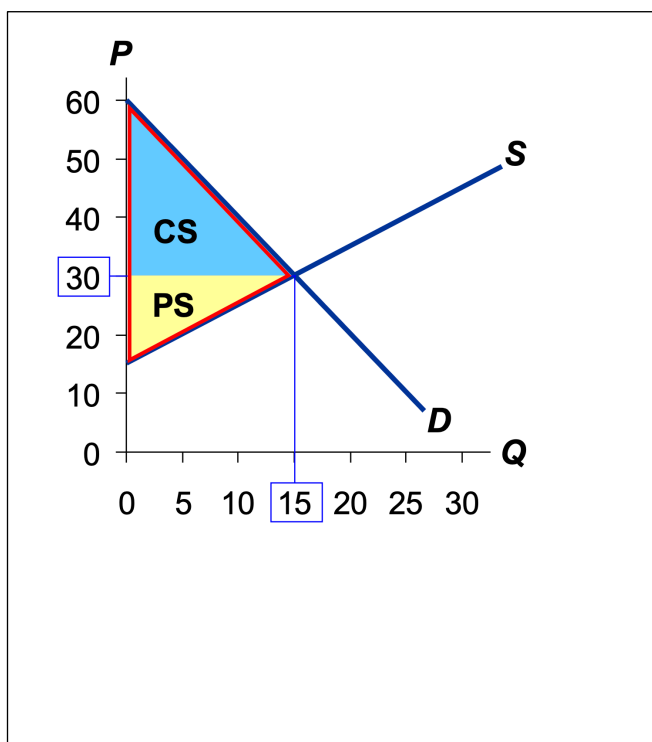
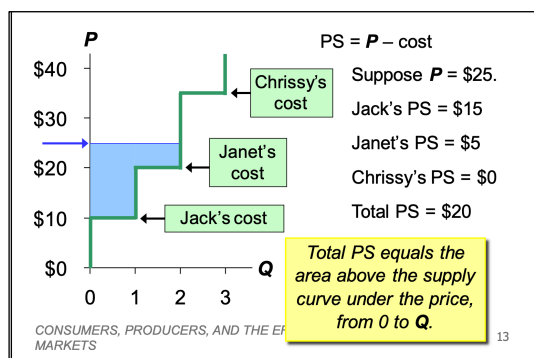
amount a seller is paid for a good minus the seller's cost

$$PS = P - \text{cost}$$

3. Total Surplus (TS):

$$TS = CS + PS$$

total gains from trade in a market



Module 52

Explicit vs. Implicit costs:

- Explicit Cost:** the cost that involves the actual layout money.
→ Actual money
- Implicit Cost:** the measure of the value of the forgone benefits.
→ Opportunity costs, tradeoffs, etc.

They add together equals to the total opportunity cost!

Economic vs. Accounting profit:

- Economic profit:** business's revenue minus the opportunity cost
 - It is often less than the accounting profit.
 - $R - \text{explicit cost} - \text{OC}$
- Accounting profit:** the business's revenue minus the explicit costs and depreciation
 - $R - \text{explicit cost}$

Capital: the value of its assets

- Ex: equipment, buildings, tools, inventory, and financial assets.

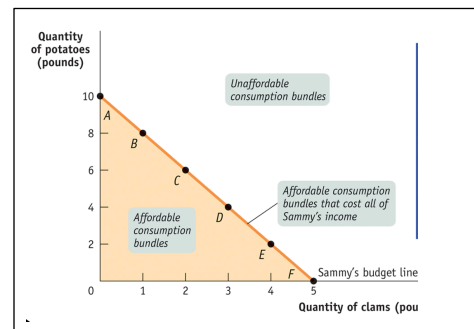
Normal profit: A zero economic profit

- just enough to keep the resources it employs in its current activity

Module 53:

Production possibility curve (PPC):

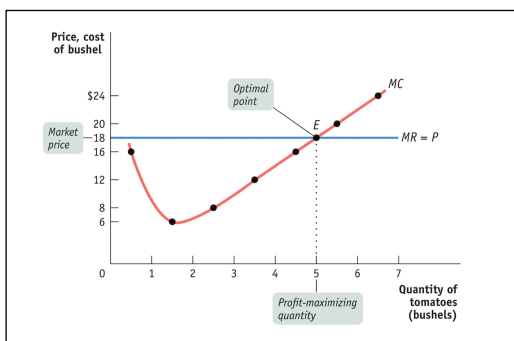
- Capital, land labor, interpreter
- Efficiency:
 - inside the curve: inefficient
 - on the curve: efficient
 - outside the curve: Impossible



Marginal Utility: additional utility from spending one more dollar (unit) on that good or service.

- Marginal utility of one unit / Price of one unit of the good = marginal utility per unit
- $MU_{\text{good}} / P_{\text{good}}$

$$\sum \text{Marginal Utility} = \text{Total Utility}$$



- **optimal output rule:** profit is maximized by producing the quantity of output at which the marginal cost of the last unit produced is equal to its marginal revenue.

- **firm's optimal output rule:** the firm's profit is maximized by producing the quantity of output

at which the marginal cost of the last unit produced is equal to the market price.

Module 54-55:

Fixed vs. Variable input:

Fixed input: input, quantity is fixed for a period of time and cannot be varied.

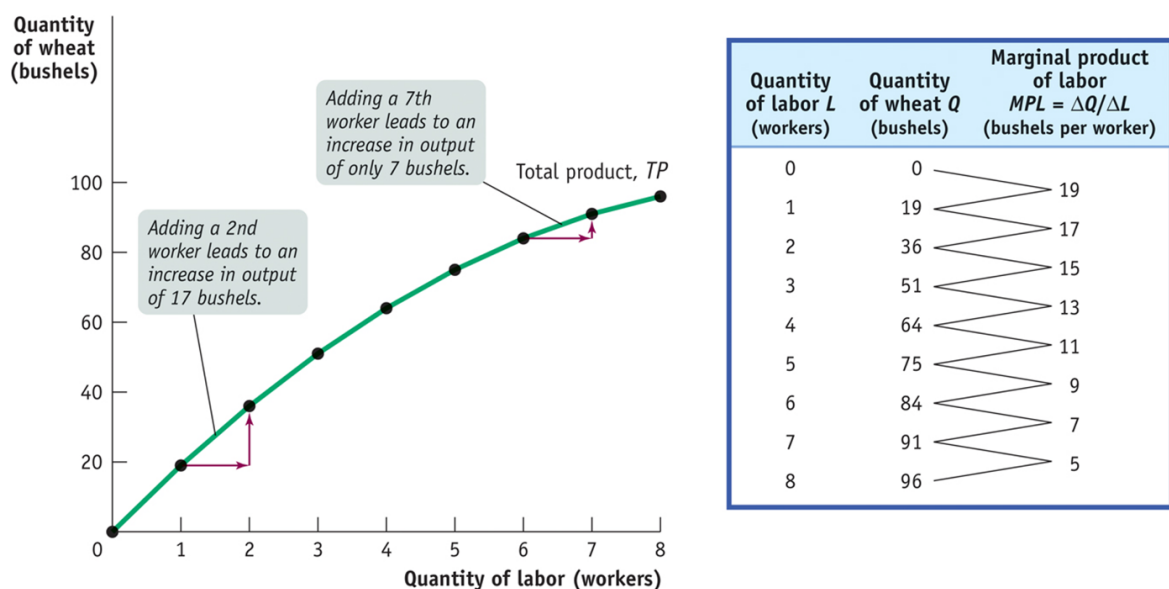
Variable input: input, quantity the firm can vary at any time

Long vs. Short run:

Long run: time period in which all inputs can be varied

Short run: time period in which at least one input is fixed

Total Product curve: how Q of output depends on G of a variable input, for a given fixed input



Marginal Product: additional quantity of output that is produced by using one more unit input

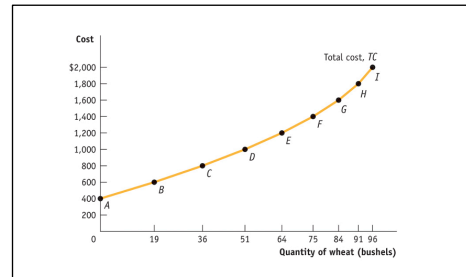
- Marginal product = $\Delta Q \text{ output} / \Delta Q \text{ input (labor)}$

- $MPL = \Delta Q / \Delta L$
- Marginal product diminishes as input increases

Total Cost: sum of fixed cost and variable cost of producing that Q of output.

- $TC = FC + VC$
- Total cost curve becomes steeper as more output is produced

→ diminishing returns



Marginal Cost = $\Delta \text{total cost} / \Delta Q \text{ output}$

Average total cost (average cost): total cost divided by Q of output

→ $ATC = TC / Q$

U-shaped: ATC falls as low levels of output, then rises

Average fixed cost: fixed cost per unit of output

→ $AFC = FC / Q$

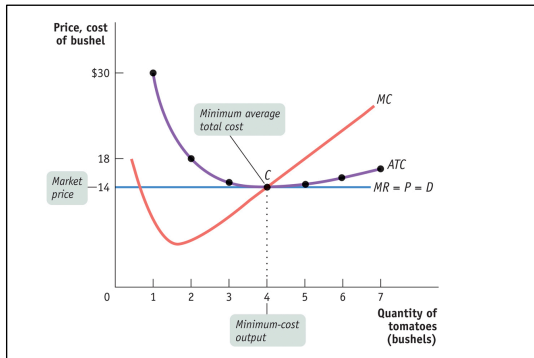
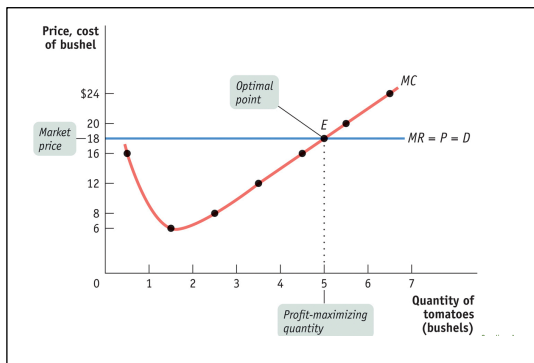
Cost:

- **Fixed cost:** the cost that does not depend on the quantity of output produced
- **Variable cost:** the cost that depends on the quantity of output produced

Module 56-58:

Marginal revenue: curve shows how marginal revenue varies as output varies

- If $TR > TC$, the firm is **profitable**.



- If $TR = TC$, the firm **breaks even**.
- If $TR < TC$, the firm **incurs a loss**.

1. **Break-even price:** market price at which it earns zero profits.
2. market price $>$ minimum ATC
→ profitable.
3. market price = minimum ATC
→ breaks even.
4. market price $<$ minimum ATC
→ unprofitable.

Market Structure:

1. Perfect competition:

- Firms are price-takers
- Free entry and exit
- No barriers

2. Monopoly:

- no close substitutes
- industry controlled by a monopolist
- barriers to entry:
 - a. Control of a Scarce Resource or Input
 - b. Economies of scale
 - c. Technological superiority
 - d. Government created barriers

3. Oligopoly:

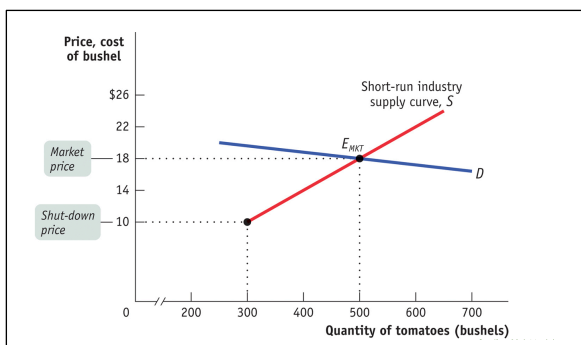
- small number of large firms
- Often related to game theory
- Characteristics of an oligopoly:
 - c. a few large firms
 - d. barriers to entry
 - e. interdependence

4. Monopolistic Competition:

- Large Numbers
- Differentiated Products
- Free Entry and Exit in the Long Run

Characteristic	Perfect Competition	Monopolistic Competition	Oligopoly	Monopoly
Number of firms	Many	Many	Few	One
Product differentiation	No	Yes	Yes or No	No
Control over price	None	Within narrow limits	Varies, depending on the degree of mutual interdependence and collusion	Considerable, limited by consumers' willingness to pay
Barriers to entry	None	Few if any	Considerable	Prohibitive
Example	Agriculture	Restaurants	Airlines	Utilities

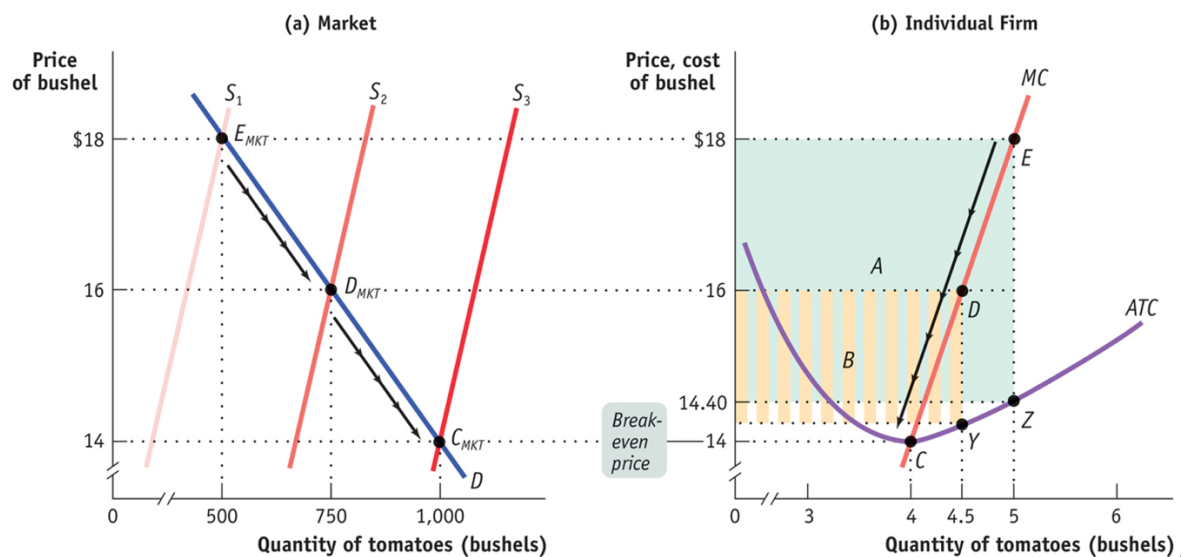
Module 60-62:



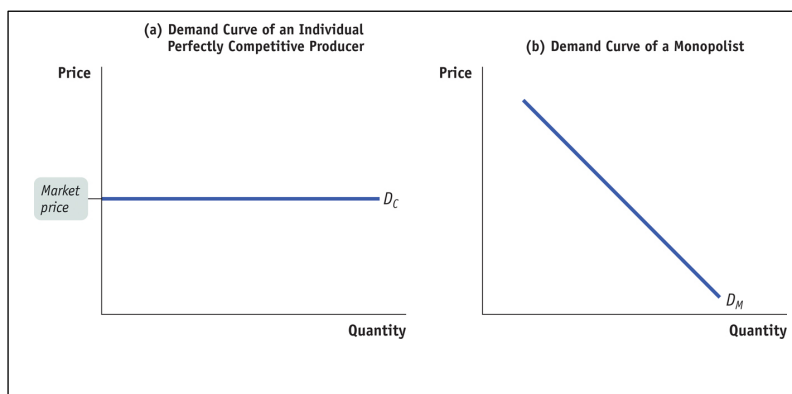
- The industry supply curve shows the relationship between the price of goods and the total output of the industry as a whole.
- Short-run market equilibrium: quantity supplied equals quantity demanded

Long-run industry supply curve: industry supply curve given sufficient time for entry into and exit from the industry.

long-run market equilibrium: quantity supplied equals the quantity demanded

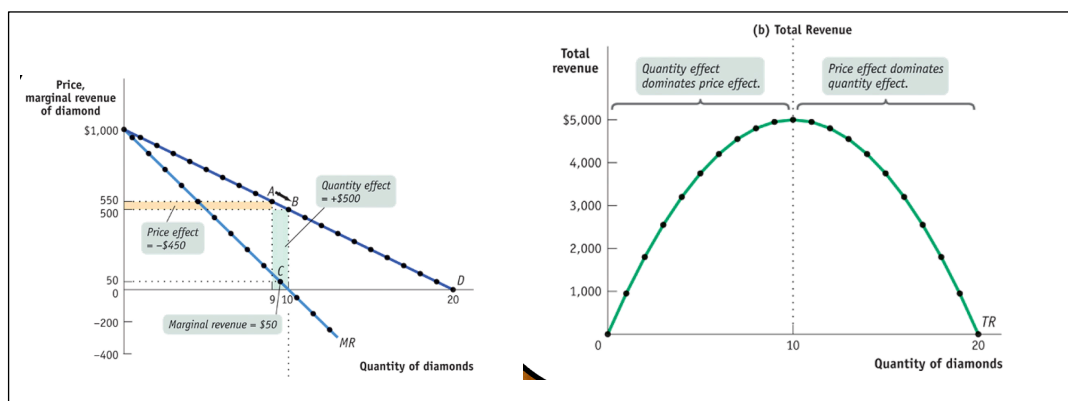


Demand curve in a monopoly:

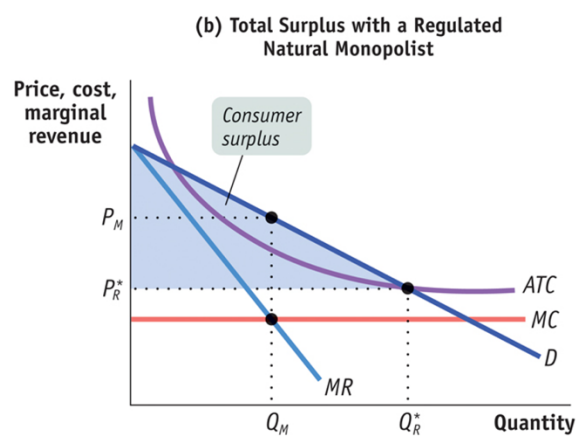
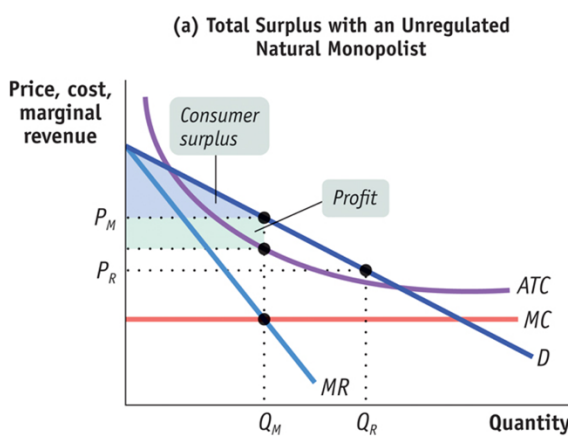
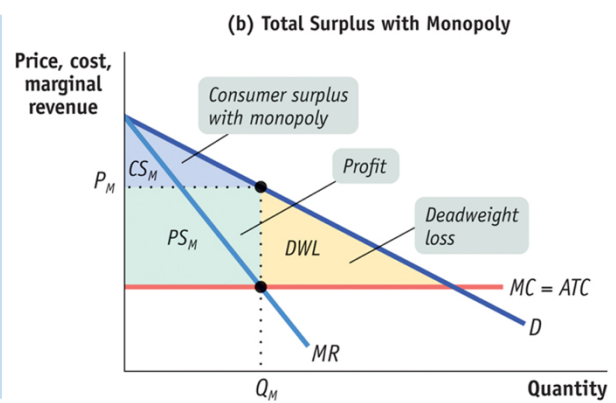
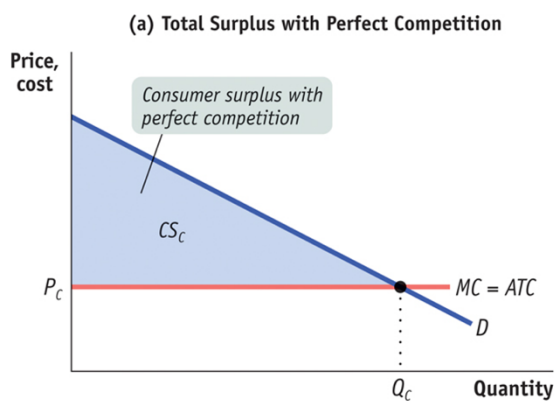
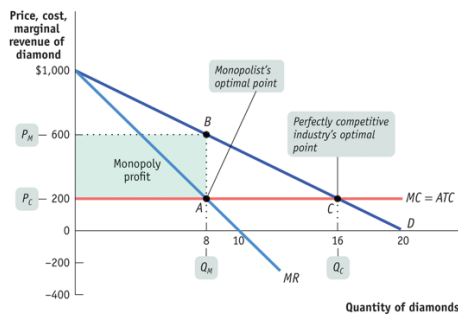


Quantity effect: One more unit is sold, increasing total revenue by the price at which the unit is sold.

Price effect: To sell the last unit, the monopolist must cut the market price on all units sold. This decreases total revenue.



A firm stops producing when $MC=MR$



Labor Market

1. Perfectly competition labor Market

2. Monopsony

MFC= Marginal Factor Cost

MRP: Marginal revenue product

In perfectly competition, the labor market set a price, and this price is the MFC for individual firms. The Demand for Labor in individual firms, which is the MRP of it, is sloping downward. The firm are price taker and they will hire at the quantity where the MRP meet the MFC.

In Monopsony, the firm is the only firm in a labor market. This cause it to control the wage. It produces at the quantity where $MFC = \text{Demand (MRP)}$, and the price is the intersection point between Supply curve and the quantity produced.

